Antihypertensive Agents

Formulary Monograph: Classes, Dosing, and Contraindications

Illustrative / synthetic formulary document created for training and demonstration purposes (e.g., testing retrieval-augmented generation systems). Not a substitute for an approved drug reference; always consult current, authoritative prescribing information before clinical use.

1. Purpose

This monograph summarizes the major classes of antihypertensive medications available on the institutional formulary, including standard adult dosing, key contraindications, monitoring requirements, and clinically significant interactions. It is intended as a quick-reference resource for prescribers and pharmacists.

2. ACE Inhibitors

Mechanism: Inhibit angiotensin-converting enzyme, reducing angiotensin II formation and aldosterone release, resulting in vasodilation and reduced sodium/water retention.

Agent	Starting Dose	Usual Range	Max Dose	Frequency
Lisinopril	10 mg once daily	10–40 mg/day	80 mg/day	Once daily
Enalapril	5 mg once daily	10–40 mg/day	40 mg/day	Once or twice daily
Ramipril	2.5 mg once daily	2.5–10 mg/day	20 mg/day	Once daily

Key contraindications: pregnancy (all trimesters), bilateral renal artery stenosis, history of angioedema with ACE inhibitor use, concurrent use with aliskiren in patients with diabetes.

Monitoring: serum creatinine/eGFR and potassium at baseline, 1–2 weeks after initiation or dose change, then periodically. A rise in creatinine of up to 30% from baseline may be acceptable; larger rises warrant reassessment.

Notable interactions: potassium-sparing diuretics and potassium supplements (hyperkalemia risk); NSAIDs (reduced antihypertensive effect, renal risk); lithium (increased lithium levels).

3. Angiotensin Receptor Blockers (ARBs)

Mechanism: Block the AT1 receptor, preventing the vasoconstrictive and aldosterone-stimulating effects of angiotensin II. Generally used as an alternative for patients intolerant of ACE inhibitors due to cough.

Agent	Starting Dose	Usual Range	Max Dose	Frequency
Losartan	50 mg once daily	50–100 mg/day	100 mg/day	Once or twice daily
Valsartan	80 mg once daily	80–320 mg/day	320 mg/day	Once daily
Telmisartan	40 mg once daily	40–80 mg/day	80 mg/day	Once daily

Contraindications and monitoring largely mirror ACE inhibitors (pregnancy, bilateral renal artery stenosis, renal function/potassium monitoring), though cough and angioedema are far less common with ARBs.

4. Calcium Channel Blockers

Mechanism: Inhibit calcium influx into vascular smooth muscle (dihydropyridines) or cardiac/vascular tissue (non-dihydropyridines), producing vasodilation and, for non-dihydropyridines, reduced heart rate and contractility.

Agent	Class	Starting Dose	Usual Range	Key Caution
Amlodipine	Dihydropyridine	5 mg once daily	2.5–10 mg/day	Peripheral edema
Nifedipine ER	Dihydropyridine	30 mg once daily	30–90 mg/day	Avoid immediate-release form
Diltiazem ER	Non-dihydropyridine	120 mg once daily	120–360 mg/day	Avoid with beta-blockers in heart block
Verapamil ER	Non-dihydropyridine	120 mg once daily	120–480 mg/day	Constipation; strong CYP3A4 inhibitor

5. Thiazide and Thiazide-Like Diuretics

Mechanism: Inhibit sodium reabsorption at the distal convoluted tubule, promoting natriuresis and volume reduction; long-term antihypertensive effect is also mediated by vasodilation.

- Hydrochlorothiazide: 12.5–25 mg once daily (max 50 mg/day)
- Chlorthalidone: 12.5–25 mg once daily (max 25 mg/day); longer half-life, often preferred for outcomes data
- Indapamide: 1.25–2.5 mg once daily

Monitoring: serum sodium, potassium, and uric acid at baseline and periodically. Use with caution in patients with gout (may precipitate flares) or significant hyponatremia.

6. Beta-Blockers

Mechanism: Block beta-adrenergic receptors, reducing heart rate, contractility, and renin release. Not first-line for uncomplicated hypertension but preferred when compelling indications exist (e.g., post-myocardial infarction, heart failure with reduced ejection fraction, certain arrhythmias).

- Metoprolol succinate (extended-release): 25–200 mg once daily
- Bisoprolol: 2.5–10 mg once daily
- Carvedilol: 3.125–25 mg twice daily (titrate slowly in heart failure)

Key cautions: avoid abrupt discontinuation (risk of rebound hypertension/tachycardia); use cautiously in reactive airway disease and decompensated heart failure; may mask hypoglycemia symptoms in diabetes.

7. Comparative Contraindication Summary

Class	Key Absolute/Relative Contraindications
ACE Inhibitors	Pregnancy, bilateral renal artery stenosis, prior angioedema
ARBs	Pregnancy, bilateral renal artery stenosis
Dihydropyridine CCBs	Severe aortic stenosis (relative), significant peripheral edema
Non-DHP CCBs	Heart failure with reduced ejection fraction, second/third-degree AV block, concurrent beta-blocker with conduction disease
Thiazides	Severe hyponatremia, symptomatic gout, sulfa allergy (relative)
Beta-Blockers	Severe bradycardia, decompensated heart failure, severe reactive airway disease

8. Additional Agent Classes

8.1 Aldosterone Antagonists (Mineralocorticoid Receptor Antagonists)

Spironolactone (12.5–50 mg once daily) and eplerenone (25–100 mg/day) are commonly used as fourth-line add-on agents in resistant hypertension. Both carry a meaningful risk of hyperkalemia, particularly when combined with an ACE inhibitor or ARB, and require periodic potassium and renal function monitoring. Spironolactone may also cause gynecomastia and menstrual irregularities due to off-target hormonal effects.

8.2 Alpha-1 Blockers

Doxazosin (1–8 mg once daily) and terazosin (1–20 mg once daily) are generally reserved for patients with coexisting benign prostatic hyperplasia. First-dose hypotension and orthostatic symptoms are notable risks, and doses should be initiated at bedtime with slow titration.

8.3 Direct Vasodilators

Hydralazine (10–100 mg, two to four times daily) and minoxidil (2.5–100 mg/day) are potent vasodilators typically reserved for resistant hypertension, generally used in combination with a beta-blocker (to blunt reflex tachycardia) and a diuretic (to counter fluid retention). Hydralazine carries a risk of drug-induced lupus at higher doses; minoxidil commonly causes hypertrichosis.

8.4 Central Alpha-2 Agonists

Clonidine (0.1–0.8 mg/day, oral or transdermal patch) acts centrally to reduce sympathetic outflow. Abrupt discontinuation can cause severe rebound hypertension, so tapering is required when stopping therapy.

9. Hypertensive Urgency and Emergency — Formulary Considerations

This monograph addresses chronic outpatient management. Severe, acute blood pressure elevation with signs of end-organ damage (hypertensive emergency) requires intravenous agents and inpatient/critical care management, which fall outside the scope of this document. Oral agents listed above are not appropriate for rapid blood pressure reduction in an emergency setting.

10. Adherence and Formulation Considerations

- Fixed-dose combination products (e.g., ACE inhibitor/thiazide, ARB/CCB) can reduce pill burden and improve adherence, and should be considered once individual component doses are established
- Extended-release formulations are generally preferred over immediate-release equivalents where available, to support once-daily dosing and smoother blood pressure control
- Generic formulations are available for most agents listed in this monograph and should be used preferentially where clinically equivalent, to support affordability and long-term adherence
- Patients transitioning between formulations of the same agent (e.g., immediate-release to extended-release) should have blood pressure reassessed within 1–2 weeks

11. General Prescribing Notes

- Combination therapy (e.g., ACE inhibitor/ARB + CCB, or ACE inhibitor/ARB + thiazide) is often required to reach blood pressure targets and may allow lower doses of each agent.
- Do not combine an ACE inhibitor with an ARB due to increased risk of hyperkalemia and renal impairment without added benefit.
- Reassess renal function and electrolytes within 1–2 weeks of any dose increase in a renin-angiotensin system blocker or diuretic.
- Consider medication cost, once-daily dosing, and pill burden to support long-term adherence.

12. Interaction Screening Summary

Combination	Effect	Recommendation
ACE-I/ARB + Potassium-	Additive hyperkalemia risk	Monitor potassium closely; avoid in renal

sparing diuretic		impairment
Non-DHP CCB + Beta-blocker	Additive negative chronotropic/inotropic effect	Avoid combination in conduction disease or reduced ejection fraction
Thiazide + Lithium	Reduced lithium clearance, risk of toxicity	Monitor lithium levels closely if combination is unavoidable
ACE-I/ARB + NSAIDs	Reduced antihypertensive effect, renal impairment risk	Use lowest effective NSAID dose for shortest duration; monitor renal function

13. Patient Counselling Points

- Take antihypertensive medications at a consistent time each day, and do not stop or skip doses even when blood pressure readings look normal
- Report symptoms such as persistent dry cough (ACE inhibitors), swelling of the ankles (dihydropyridine CCBs), or unusual muscle weakness (diuretics) to a healthcare provider
- Avoid abruptly stopping beta-blockers or centrally-acting agents such as clonidine without medical guidance, due to rebound effects
- Keep scheduled blood tests for kidney function and electrolytes, particularly after starting or increasing a renin-angiotensin system blocker or diuretic

End of document. Document ID: MED-FORM-002 (synthetic training corpus).